

15

Analysis of Covariance (ANCOVA)

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ANCOVA: Main Concepts

Now, for a controversial research topic. There is evidence to suggest that women have more brain cells per cubic centimeter, more grey matter on a percentage of total brain volume basis, and greater neuronal inter-connections than men (de Courten-Meyers, 1999; Gur et al., 1999). A lot of women would be thinking to themselves right now, “Yeah, that sounds about right.” However, there is also evidence to suggest that men and women, on average, do not differ in overall levels of intelligence (Neisser et al., 1996). Consequently, how might the difference in the brain related characteristics favoring females be reconciled with the absence of overall gender difference in intelligence? The data I collected for my honors thesis (back when having [this](#) mobile phone was a status symbol) might be able to help answer this question. Specifically, I collected data from men ($N = 40$) and women ($N = 60$) to estimate overall brain weight, body size, and intelligence. Based on independent samples t -test, the null

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hypothesis of equal brain weight means was rejected, $t = -7.14$, $p < .001$, Cohen's $d = -1.43$. Thus, it may be suggested that, on average, males have larger brains than females (males: $M = 1.47$, $SD = .16$; females $M = 1.27$, $SD = .12$). Superficially, such a result might be interpreted to help reconcile the aforementioned inconsistent findings. That is, sure, men don't have as many brain cells per cubic centimeter, as much grey matter on a percentage basis, or as many brain inter-connections as women, on average, but men, on average, have more overall brain weight/size. So, the brain related differences might even out, which is consistent with the absence of an overall difference between men and women in intelligence. Everyone is happy, right? Well, perhaps not so fast.

It needs to be acknowledged that males also had larger body sizes than females in my Honours thesis data, $t = -12.93$, $p < .001$, Cohen's $d = -2.82$. Such a result suggests that the only reason males might have, on average, larger brains than females is because they have overall larger bodies. That is, it's not just brain size on its own that one should take into consideration, as even a baby elephant has a larger brain than the typical adult human. However, it is probably not justifiable to suggest that baby elephants are more intelligent than adult humans, even though they appear to be social with other animals.¹



Thus, it would be natural to wonder whether males have larger brains than females, controlling for the differences between males and females in body size. Stated alternatively, if one could equalize males and females for body size statistically, would males continue have larger mean brain sizes? Ultimately, it is possible that males have, on average, larger brains than females, simply because they have larger overall body sizes. The analysis of covariance (ANCOVA) can help address this issue statistically.

ANCOVA is very similar to the between-subjects ANOVA. The key exception is that ANCOVA calculates adjusted group means on the basis of one or more covariates. A covariate is a variable that is correlated with either an independent variable or a dependent variable (or both). In ANCOVA, the means are adjusted, based on the association between the independent variable and the covariate, as well as the association between dependent variable and the covariate (assuming there is an association between the covariate and both the independent and dependent variables). Very typically, the association between the covariate and the independent/dependent variables is positive. The presence of such combined positive associations causes the means to regress toward each other. With respect to my honors thesis, statistically, the question was whether the male and female brain volume

¹ I can't find the copyright holder of this photo. I would provide attribution, if I could.

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means regressed toward each to such a degree that there was no longer a statistically significant difference between the means.

Prior to conducting the ANCOVA in SPSS, I have re-reported the original effect of sex on brain weight. However, I have conducted the analysis with a one-way between-subjects ANOVA, rather than an independent samples *t*-test, to help ensure easier comparability in the results. Specifically, I have used SPSS' General Linear Model menu option ([Data File: brain_size](#)) ([Watch Video 15.1: Pre-ANCOVA Between-Subjects ANOVA in SPSS](#))

Descriptive Statistics

Dependent Variable: brain_size_kgs

sex	Mean	Std. Deviation	N
Female	1.2706	.12301	60
Male	1.4740	.16159	40
Total	1.3519	.17127	100

It can be observed that the female and male groups had brain weight means of 1.27 and 1.47 kilograms, respectively. The Cohen's *d* corresponded to -1.43. Finally, as can be seen below, the null hypothesis of equal mean brain weights between males and females was rejected, $F = 50.91$, $p < .001$, $\eta^2 = .342$. Note that the General Linear Model utility in SPSS refers to the effect size as partial eta-squared, even when it is not. When the analysis is consistent with a one-way between-subjects ANOVA, the effect size reported by SPSS is definitely eta-squared, not partial eta-squared. Thus, 34.2% of the variance in brain weight was accounted for by sex. However, such an observation is not particularly interesting, as males have larger body sizes than females. The key questions are, "How much brain weight variance can be accounted for by sex, when controlling for the effects of body size. Furthermore, is the effect significant statistically?" The advantage of conducting a one-way between subjects ANOVA through the General Linear Model utility in SPSS is that you don't have to calculate eta-squared by hand.

Tests of Between-Subjects Effects

Dependent Variable: brain_size_kgs

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	.993 ^a	1	.993	50.909	.000	.342
Intercept	180.784	1	180.784	9270.455	.000	.990
sex	.993	1	.993	50.909	.000	.342
Error	1.911	98	.020			
Total	185.680	100				
Corrected Total	2.904	99				

a. R Squared = .342 (Adjusted R Squared = .335)

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Hypotheses

Null Hypothesis (H_0): Male and female brain weight means are equal, controlling for individual differences in body size.

Alternative Hypothesis (H_1): Male and female brain weight means are unequal, controlling for individual differences in body size.

ANCOVA: SPSS

In order to perform an ANCOVA in SPSS, you can use the 'General Linear Model' menu option in SPSS (**Watch Video 15.2:** [ANCOVA in SPSS](#))

Descriptive Statistics

Dependent Variable: brain_size_kgs

sex	Mean	Std. Deviation	N
Female	1.2706	.12301	60
Male	1.4740	.16159	40
Total	1.3519	.17127	100

As can be seen in the 'Descriptive Statistics' table, the means, standard deviations and sample sizes have been reported. Males have, on average, 203 grams more brain weight than females. These means are unadjusted. SPSS reports the adjusted means in a table further below.

Tests of Between-Subjects Effects

Dependent Variable: brain_size_kgs

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	1.064 ^a	2	.532	28.027	.000	.366
Intercept	165.360	1	165.3...	8715.6	.000	.989
body_size_factor	.071	1	.071	3.729	.056	.037
sex	.126	1	.126	6.652	.011	.064
Error	1.840	97	.019			
Total	185.680	100				
Corrected Total	2.904	99				

a. R Squared = .366 (Adjusted R Squared = .353)

The table above entitled 'Test of Between-Subjects Effects' is rather busy. However, you need to focus only upon the row relevant to the independent variable, which in this case corresponds to 'sex'. As can be seen above, the null hypothesis of equal (adjusted) means between males and females in brain weight has been rejected, $F(1, 97) = 6.65$, $p = .011$, partial $\eta^2 = .064$. Thus, even controlling for body size, males were associated with a larger mean brain weight than females, in this sample. Specifically, approximately 6.4% of the variance in brain

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weight was accounted for by sex, controlling for the effects of body size. Cohen (1988) did not provide effect size guidelines for ANCOVA, specifically. It is up to you whether you would use Cohen's (1988) between-subjects ANOVA guidelines as a substitute. I can see arguments for and against, in this context.

sex

Dependent Variable: brain_size_kgs

sex	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
Female	1.301 ^a	.024	1.254	1.349
Male	1.428 ^a	.032	1.363	1.492

a. Covariates appearing in the model are evaluated at the following values: body_size_factor = .0000000.

Finally, as can be seen in the table above, the male and female adjusted brain weight means were 1.428 and 1.301, respectively. Thus, including the body size covariate in the analysis, the mean difference between males and females in terms of brain size was reduced to .127, or 127 grams ($1.428 - 1.301$). Recall that the original means were associated with a mean difference of .203, or 203 grams ($1.474 - 1.271$). Given that $203 - 127 = 76$, it may be suggested that body size accounted for 76 grams of the original brain weight difference effect. Thus, one of the reasons males and females have about equal levels of overall intelligence, despite the neurophysiological advantages associated with females, is that males have, on average, greater brain weight. So, we can all breathe a sigh of relief. Males and females are essentially equal with respect to overall intelligence, and how that equality emerges appears to be via two different neurophysiological routes.

Effect Size

SPSS provides the option to estimate partial eta-squared, which is a perfectly acceptable approach to the reporting of an effect size in the case of an ANCOVA. Less well-known is that it is possible to calculate Cohen's d , as well, based on the adjusted means and adjusted standard deviations. Unfortunately, SPSS does not report the adjusted standard deviations. It only reports the adjusted standard errors (in addition to the adjusted means). However, the adjusted standard deviations can be calculated simply enough by multiplying the standard errors by their respective square root of N . Thus, as can be seen in Table C15.1, the adjusted standard deviations for females and males were .186 and .202, respectively (**Watch Video 15.3: [ANCOVA-Based Adjusted Standard Deviations in SPSS](#)**):

For comparison, the females and males unadjusted SDs were .123 and .162, respectively. Thus, the standard deviations increased as a consequence of including the body size covariate in the analysis. The smaller numerical difference between the means and the

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larger standard deviations are a direct cause of the smaller effect size obtained in the ANCOVA (partial $\eta^2 = .064$), in comparison to the between-subjects ANOVA ($\eta^2 = .342$).

Table C15.1. Calculations to Obtain Adjusted Standard Deviations

	SE	N	$\sqrt{N}$	SD_A
Females	.024	60	7.746	.186
Males	.032	40	6.324	.202

Note. SD_A = adjusted standard deviation; $SD_A = SE \cdot \sqrt{N}$

In order to calculate Cohen's adjusted d , simply use the original Cohen's d formula with the adjusted means and adjusted standard deviations.

$$\text{adjusted Cohen's } d = \frac{\bar{X}_{A1} - \bar{X}_{A2}}{(SD_{A1} + SD_{A2})/2} \quad (1)$$

$$\text{adjusted Cohen's } d = \frac{1.301 - 1.428}{(.186 + .202)/2} = \frac{-.127}{.194} = -.655$$

Thus, the adjusted Cohen's d was estimated at -.655, which is between a medium and large effect size (Cohen, 1988). Consequently, it may be suggested that the female group was associated with a .655 of an adjusted standard deviation smaller brain size than males. On the basis of an independent samples t -test (or ANOVA), the unadjusted Cohen's d estimate corresponded to -1.43. Thus, the effect size reduced in magnitude from -1.43 to -.655, based on the inclusion of the body size covariate. If you wish, you could calculate the slightly more accurate Hedge's g , in this context (see Chapter 6).

To summarize, although females appear, as a group, to have brain related characteristics that afford them some advantages, males also appear to have at least one neurophysiological advantage, even controlling for body size. Prior to interpreting the result, however, it is necessary to test a key statistical assumption associated with the ANCOVA that is relatively unique, as I describe next.

Assumptions

The assumptions associated with the ANCOVA are the same as those associated with the between-subjects one-way ANOVA (see Chapter 7). Additionally, the ANCOVA has a unique assumption. It is known as **homogeneity of regression slopes**. The assumption implies that the association between the covariate and the dependent variable is equal across all groups. Recall that the ANCOVA produces adjusted means and standard deviations. Those adjusted means and standard deviations are derived from a common (pooled) slope (unstandardized beta weight). Consequently, the unstandardized beta weights need to be equal across all groups, within sampling fluctuations, in order to apply the conventional ANCOVA.

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Fortunately, empirical simulation research has found ANCOVA to be fairly robust to violations of the assumption of homogeneity of regression slopes. For example, Wu (1984) found that when the standardized beta weights are different by $|\leq .40|$ or less, the ANCOVA is robust to violations of the homogeneity of regression slopes. If the sample sizes are unequal, the standardized beta weights can also differ by as much as $|\leq .40|$. However, the homogeneity of variance (Levene's) assumption has to be satisfied, if the sample sizes are unequal. If the sample sizes are equal, the homogeneity of variance (i.e., Levene's) assumption does not have to be satisfied, but the $|\leq .40|$ rule for the regression slopes still applies. If the homogeneity of regression slopes is severely violated (difference greater than $|\leq .40|$ in standardized beta weights), then the more sophisticated option of conducting an ANCOVA on separate slopes is known to be robust (Wu, 1984). Unfortunately, few (if any) programs offer the option of ANCOVA via separate slopes. An alternative to the separate slopes approach is the picked-points analysis (Huitema, 2011), which I demonstrate in the Advanced Topics section of this chapter.

Even the separate slopes ANCOVA assumes the sample sizes are equal, if the homogeneity of variance (Levene's) is violated. If you violate the assumption of homogeneity of variance, and you have unequal sample sizes, and you are keen on keeping N as large as possible, you could argue that ANCOVA is known to be a conservative test, when the larger standard deviation is associated with the larger sample size (Rheinheimer & Penfield, 2001). However, if the larger standard deviation is associated with the smaller sample size, you should consider making your sample sizes equal across groups (using random sampling; see Chapter 9). Finally, just like the between-subjects ANOVA, ANCOVA is known to be robust to violations of normality. Rheinheimer and Penfield (2001) tested the robustness of ANCOVA up to non-normal distributions of skew = $|\leq 1.5|$ and kurtosis and of $|\leq 3.0|$. They found ANCOVA was almost always robust up to such levels of non-normality.

Testing Assumptions: Normality

First, the assumption of normality can be investigated by splitting the data file across groups and selecting skew and kurtosis within the Frequencies utility (**Watch Video 15.4: [ANCOVA Normality and Homogeneity of Variance in SPSS](#)**). I obtained the following result:

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Statistics				
sex			brain_size_kgs	body_size_factor
Female	N	Valid	60	60
		Missing	0	0
	Skewness		.088	-.649
	Std. Error of Skewness		.309	.309
	Kurtosis		-.128	1.323
	Std. Error of Kurtosis		.608	.608
Male	N	Valid	40	40
		Missing	0	0
	Skewness		-.968	.331
	Std. Error of Skewness		.374	.374
	Kurtosis		1.594	-.484
	Std. Error of Kurtosis		.733	.733

As be seen above, the skew and kurtosis levels were less than |1.5| and |3.0|, respectively. Consequently, the data were determined to be sufficiently normal for the purposes of conducting an ANCOVA. Next, the assumption of homogeneity of variance can be tested through the General Linear Model utility. I obtained the following result:

Levene's Test of Equality of Error Variances^a

Dependent Variable: ...

F	df1	df2	Sig.
2.952	1	98	.089

Tests the null hypothesis that the
error variance of the dependent
variable is equal across groups.

a. Design: Intercept +
body_size_factor + sex

As can be seen above, the assumption of homogeneity of variances has been satisfied, $F(1, 98) = 2.95$, $p = .089$. This is a particularly favorable outcome, given that the sample sizes were unequal.

Testing Assumptions: Homogeneity of Regression Slopes

The homogeneity of regression slopes assumption is tested via an interaction hypothesis. If the association between the independent variable and the dependent variable is found to be moderated (influenced) by the covariate, the assumption of homogeneity of regression slopes is considered to have been violated. In practice, the interaction is tested by the creation of a product term between the independent variable and the covariate. As per the ANCOVA, the homogeneity of regression slopes assumption can be tested via the General Linear Model utility (**Watch Video 15.5: [Test Homogeneity of Regression Slopes in SPSS](#)**). When I did so in SPSS, I obtained the following result:

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Tests of Between-Subjects Effects

Dependent Variable: brain_size_kgs

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	1.082 ^a	3	.361	19.011	.000	.373
Intercept	60.730	1	60.730	3200.400	.000	.971
sex	.123	1	.123	6.483	.012	.063
body_size_factor	.085	1	.085	4.458	.037	.044
sex * body_size_factor	.019	1	.019	.986	.323	.010
Error	1.822	96	.019			
Total	185.680	100				
Corrected Total	2.904	99				

a. R Squared = .373 (Adjusted R Squared = .353)

As can be seen in the SPSS table entitled 'Tests of Between-Subjects Effects', the 'sex*body_size_factor' interaction term was not significant statistically, $F(1,96) = .99$, $p = .323$, partial $\eta^2 = .010$. Therefore, the assumption of homogeneity of regression slopes can be assumed to be satisfied. It should be emphasized that the test of the assumption of homogeneity regression slopes described here is dependent upon sample size. That is, if the sample size is large, a statistically significant interaction term may be observed, even though the difference in the regression slopes across groups is relatively trivial. Similarly, if the sample size is small, the test of the interaction term will lack power. Recall that Wu's (1984) simulation study demonstrated that the ANCOVA is robust up to a difference in standardized regression coefficients equal to .40. Therefore, it is arguably useful to estimate separate bivariate regressions for each of the groups. When I did so in SPSS, I obtained the following result:

Coefficients^a

sex	Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
			B	Std. Error	Beta		
Female	1	(Constant)	1.322	.027		48.741	.000
		body_size_factor	.077	.034	.288	2.287	.026
Male	1	(Constant)	1.446	.044		32.711	.000
		body_size_factor	.028	.036	.124	.768	.447

a. Dependent Variable: brain_size_kgs

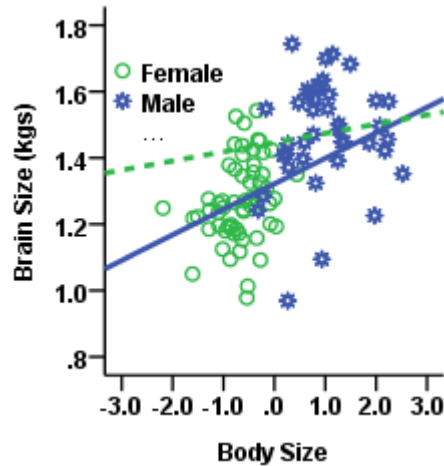
As can be seen in the SPSS table entitled 'Coefficients', the standardized regression slopes for the females and males were estimated at .288 and .124, respectively, which amounts to a difference of .164. As the difference of .164 is less than |.40|, we can assume that the homogeneity of regression slopes assumption has been sufficiently satisfied for the purposes of conducting a regular ANCOVA (**Watch Video 15.6: [Standardized Beta Weights Across Groups](#)**).

It is also possible to create a scatter plot to evaluate the homogeneity of regression slopes assumption. As can be seen in Figure C15.1, the male and female lines of best fit were

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different (Female line: broken; Male line: solid). In fact, they intersect where the body size factor approaches approximately 2.5. In my opinion, this is problematic, however, the recommended statistical test to evaluation homogeneity of regression slopes was not significant statistically. Consequently, the ANCOVA results can be interpreted.

Figure C15.1. Homogeneity of Regression Slopes Scatter Plot



What ANCOVA Does Not Assume

On this topic, I suspect my point of view will run afoul with some experts in the field. Specifically, several sources state that ANCOVA assumes that the covariate correlates only with the dependent variable (see Miller & Chapman, 2001, for example). Thus, the covariate cannot be found to correlate with the independent variable. Stated alternatively, the assumption implies that there cannot be any differences between the groups on the covariate. In the example above, the groups differed on the covariate. Specifically, males were found to have larger body sizes than females. Therefore, based on a strict interpretation of the assumption described above, many would argue that the data cannot be submitted to an ANCOVA (Miller & Chapman, 2001, and references therein). However, in my opinion, such a position is untenable, at least for non-experimental designs.² My position is based on the fact that ANCOVA and multiple regression are fundamentally identical analyses. Consequently, if the covariate cannot correlate with the independent variable in ANCOVA, independent variables cannot be inter-correlated in multiple regression: virtually no one would ever take such an argument seriously.

To prove my point of view accessibly, I conducted a multiple regression. Specifically, I regressed brain size (dependent variable) onto sex (independent variable 1) and body size

² I deal with the issue in the context of experimental designs further below.

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(independent variable 2). I obtained the following result (**Watch Video 15.7: [ANCOVA & multiple regression are the same? Yes](#)**):

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	1.301	.024		54.402	.000
sex	.126	.049	.363	2.579	.011
body_size_factor	.046	.024	.271	1.931	.056

a. Dependent Variable: brain_size_kgs

As can be seen above, the sex independent variable was associated with a statistically significant effect: $t = 2.579$, $p = .011$. In the ANCOVA, recall the sex main effect was associated with an $F = 6.652$, $p = .011$. Thus, identical p -values. A t -value can be converted into an F -value by squaring it: $2.579^2 = 6.65$. This demonstrates that multiple regression and ANCOVA are fundamentally the same analysis (see Rutherford, 2011, for formal exposition). Additionally, based on the multiple regression results above, the unstandardized slope associated with the sex variable was estimated at .126, which is identical to the difference between the adjusted means: 126 grams of brain weight! Thus, a unit increase in sex (from female to male) was associated with a 126 gram increase in brain size (on average), controlling for the effects of body size on both sex and brain size. Again, the point I made here is that any argument made against the use of ANCOVA would have to apply to multiple regression. To drive the point further, an alternative way to analyze the sex, brain size, body size data is to perform a partial correlation (**Watch Video 15.8: [ANCOVA and partial correlation are the same? Yes](#)**):

Correlations

Control Variables			sex	brain_size_kgs
body_size_factor	sex	Correlation	1.000	.253
		Significance (2-tailed)	.	.011
		df	0	97
	brain_size_kgs	Correlation	.253	1.000
		Significance (2-tailed)	.011	.
		df	97	0

As can be seen above, the correlation between sex and brain size was $r = .253$, $p = .011$, controlling for the effects of body size on sex and body size. The squared partial correlation was equal to .064, which is identical to the partial eta-squared estimated from the ANCOVA. Again, this is not a coincidence (Rutherford, 2011). Therefore, although you can find sources that criticize the use of ANCOVA when the groups differ on the covariate, these same sources curiously say nothing about multiple regression or partial correlation in this context –

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analyses which are fundamentally identical to ANCOVA. I doubt such an argument would get published in a peer-reviewed publication. If you recall the chapter on multiple regression, when X and Y are inter-related and a third variable is related to only Y , one has a potential suppressor situation. If we could only ever use multiple regression, partial correlation and ANCOVA when we have a suppressor situation, we would rarely run any multivariate analyses.

The close correspondence between ANCOVA and multiple regression has an implication for how multiple regression should be performed. That is, the assumption of homogeneity of regression slopes is often discussed in the context of ANCOVA, however, it appears to be rarely, if ever, discussed in the context of multiple regression with a dichotomous independent variable. That is, when a dichotomous independent variable (X_1) is included in a multiple regression model, researchers rarely, if ever, test whether the association between X_2 and Y (dependent variable) is the same across the two X_1 groups. Arguably, they should. The implications would be expected to extend to designs which include nominal independent variables with more than two groups.

Advanced Topics

ANCOVA: Picked-Points Analysis

When the homogeneity of regression slopes assumption is violated appreciably (difference in standardized beta weights greater than .40), the picked-points approach to ANCOVA is a viable alternative (Huitema, 2011). (forthcoming)

ANCOVA versus Mixed-Design ANOVA

The classic experimental design consists of a between-subjects factor (treatment and control) and a within-subjects factor (pre-treatment and post-treatment). Often, a researcher hypothesizes that the treatment group will evidence a greater change in the dependent variable, in comparison to the control group. As described in a different chapter (Chapter 11), the mixed-design ANOVA is especially attractive to test such a hypothesis. Specifically, the interaction term associated with the mixed-design ANOVA.

Unfortunately, many researchers have developed the habit of applying an ANCOVA to the classic experimental design. Specifically, they apply an ANCOVA with the between-subjects factor as the independent variable, the post-treatment scores as the dependent variable, and the pre-treatment scores as the covariate. They are under the impression that ANCOVA is particularly useful in such a design, when the groups differ on the pre-treatment scores, as the ANCOVA will adjust the means in such a way that there is no longer the observed pre-treatment effect on the post-treatment scores. I believe this is a mistaken impression. Allow me to illustrate with some data provided by Gollwitzer, Christ, and Lemmer (2014).

The data are consistent with a classical experimental design: one within-subjects factor with two levels (time 1 and time 2) and one between-subjects factor with two levels (condition 1 and condition 2). I performed a mixed-design ANOVA on the data and obtained the following key result ([Data File: data lords paradox short](#)) (**Watch Video 15.9: [Mixed-Design Versus ANCOVA](#)**):

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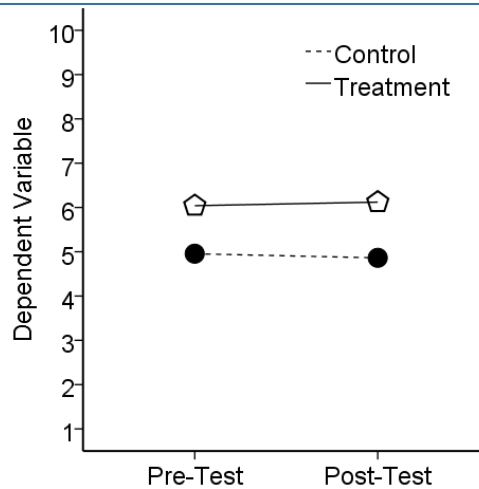
Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Sphericity Assumed	.001	1	.001	.001	.975	.000
	Greenhouse-Geisser	.001	1.000	.001	.001	.975	.000
	Huynh-Feldt	.001	1.000	.001	.001	.975	.000
	Lower-bound	.001	1.000	.001	.001	.975	.000
factor1 * condition	Sphericity Assumed	.223	1	.223	.220	.641	.004
	Greenhouse-Geisser	.223	1.000	.223	.220	.641	.004
	Huynh-Feldt	.223	1.000	.223	.220	.641	.004
	Lower-bound	.223	1.000	.223	.220	.641	.004
Error(factor1)	Sphericity Assumed	58.777	58	1.013			
	Greenhouse-Geisser	58.777	58.000	1.013			
	Huynh-Feldt	58.777	58.000	1.013			
	Lower-bound	58.777	58.000	1.013			

As can be seen above, there was no statistically significant interaction, $F(1, 58) = .22$, $p = .641$, partial $\eta^2 = .004$. The lack of effect can be appreciated by an examination of the corresponding plot of the means:

Figure C15.2. Plot of Mixed-Design ANOVA Means



Finally, the between-subjects Cohen's d values at pre-test and post-test were .76 and .89, respectively, underscoring, again, the lack of a statistically significant interaction effect. That is, the difference in the Cohen's d values was only $|.13|$. It should be case closed, right?

Unfortunately, many researchers become concerned by the observation that the two groups were not equal on the dependent variable at pre-test (see Figure C15.2). Consequently, they apply an ANCOVA to the data such that the post-test scores are used as the dependent variable and the pre-test scores are used as the covariate. The group variable is used as the independent variable. The argument is that the inclusion of the pre-test scores as a covariate

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will nullify the observed differences in the means at pre-test. This is simply not the case, necessarily. When I applied an ANCOVA to the same data depicted in Figure C15.2 above, I obtained the following results:

Tests of Between-Subjects Effects

Dependent Variable: obs_t2

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	52.829 ^a	2	26.415	17.147	.000	.376
Intercept	27.808	1	27.808	18.051	.000	.241
obs_t1	29.199	1	29.199	18.955	.000	.250
condition	6.696	1	6.696	4.346	.042	.071
Error	87.807	57	1.540			
Total	1977.703	60				
Corrected Total	140.637	59				

a. R Squared = .376 (Adjusted R Squared = .354)

As can be seen above, the between-subjects 'condition' effect was statistically significant, $F(1,57) = 4.346$, $p = .042$, partial $\eta^2 = .071$. Furthermore, the adjusted means were estimated at 5.150 and 5.868 and corresponded to a fairly sizable Cohen's $d = .718/(1.29+1.28/2) = .559$.

condition

Dependent Variable: obs_t2

condition	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
control	5.150 ^a	.244	4.662	5.639
treatment	5.868 ^a	.227	5.414	6.323

a. Covariates appearing in the model are evaluated at the following values: obs_t1 = 5.5333.

Why did the ANCOVA yield a statistically significant result? It is because the covariate did not adjust for the difference in the pre-test means; at least not sufficiently, as I discuss in more detail in the video. The ANCOVA is trying to make the adjustment on the basis of the correlation between the group variable and the continuously scored time 1 variable. However, there are limitations with such a point-biserial correlation. Ultimately, I can't think of a reason to ever use ANCOVA in a within-subjects design. The mixed-design ANOVA is a much more attractive option.

ANCOVA versus Ratios

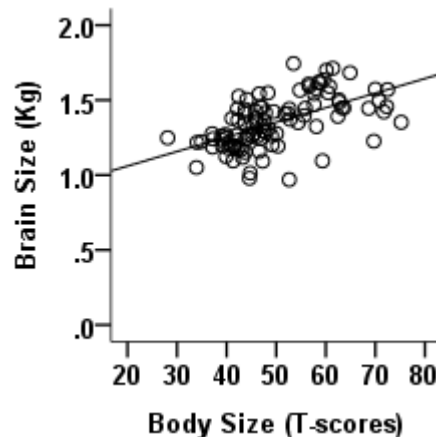
Many researchers tend to create ratios for the purposes of scaling their dependent variables, when a "nuisance" variance is correlated with the dependent variable of interest. In the context of sex and brain size study, some researchers would be inclined to divide each brain size score by the total body size score, in order to "equalize" them. Then, a regular

independent samples *t*-test or between-subjects ANOVA would be performed on the ratio scores. Such a procedure cannot be recommended, because it rests upon the assumption of isometry: a perfectly linear association between the dependent variable and the control variable that passes through the origin, i.e., intercept = 0 (Packard & Boardman, 1988). Based on a re-examination of 79 sets of field data, Packard and Boardman (1988) found that the ratio approach yielded incorrect results 28% of the time.

In addition to the very unlikely assumption of isometry, I believe a major drawback associated with the use of ratios is that the dependent variable will be rendered uninterpretable. For example, the ratio means reported by the ANOVA will be associated with an indecipherable scale. By contrast, if the dependent variable is measured on an interpretable scale to begin with, the ANCOVA will yield means and, importantly, adjusted means that are also interpretable. Consequently, on both statistical and interpretation grounds, the ANCOVA is superior to the ANOVA on ratios method.

To demonstrate the effect of using ratios on the sex, brain size, and body size data, I created ratios via the division of brain size (in kgs) by body size (transformed into T-scores to insure positive values). As can be seen in Figure C15.3, the isometry assumption was violated, as the regression line intersected with the Y-axis at a point that was not the origin (i.e., zero).

Figure C15.3. Scatter Plot with Intercept Intersecting the Y-Axis



Furthermore, as can be seen in the SPSS table entitled 'Descriptive Statistics', the females and males were associated with "brain size ratio adjusted means" of .030 and .025, respectively. Thus, numerically, the females were associated with larger brain sizes than males, which is in direct contrast to the results obtained from the ANCOVA.

CHAPTER 15: ANCOVA

Descriptive Statistics

Dependent Variable: brain_size_ratios

sex	Mean	Std. Deviation	N
Female	.030	.004	60
Male	.025	.004	40
Total	.028	.004	100

Tests of Between-Subjects Effects

Dependent Variable: brain_size_ratios

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	.001 ^a	1	.001	37.588	.000	.277
Intercept	.071	1	.071	5019.807	.000	.981
sex	.001	1	.001	37.588	.000	.277
Error	.001	98	1.417E-5			
Total	.079	100				
Corrected Total	.002	99				

a. R Squared = .277 (Adjusted R Squared = .270)

Finally, as can be seen in the SPSS table entitled 'Tests of Between-Subjects Effects', the sex main effect was statistically significant, $F(1, 98) = 37.59$, $p < .001$, $\eta^2 = .277$, which suggests that females have statistically significantly larger brains than males. Again, the ratio method of controlling is not an acceptable approach to data analysis. Consequently, I would place greater confidence in the ANCOVA results.

ANCOVA versus Residuals

An approach researchers sometimes take to evaluate hypotheses relevant to the test of the difference between two or more means is to first obtain the residuals derived from regressing the dependent variable onto the covariate. Then, conduct a regular independent sample t -test or between-subjects ANOVA on the residuals. Although this approach may appear appealing intuitively, it does not yield accurate results (García-Berthou, 2001; Maxwell, Delaney, & Manheimer, 1985). To demonstrate, I conducted an ANOVA on the residuals for the sex differences in brain weight data. First, I residualized brain weight onto body size and saved the unstandardized residuals. Then, I conducted a between-subjects ANOVA on the residuals with SPSS and obtained the following results. As can be seen in the SPSS table entitled 'Tests of Between-Subjects Effects' table, the test of the difference between male and female brain weight residual means was not significant statistically, $F(1, 98) = 2.12$, $p = .148$, $\eta^2 = .021$. Such a result is in contrast to the statistically significant result obtained from the corresponding ANCOVA.

CHAPTER 15: ANCOVA

Tests of Between-Subjects Effects

Dependent Variable: brain_size_residuals

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	41723.550 ^a	1	41723.550	2.124	.148	.021
Intercept	1668.942	1	1668.942	.085	.771	.001
sex	41723.550	1	41723.550	2.124	.148	.021
Error	1924854.54	98	19641.373			
Total	1966578.09	100				
Corrected Total	1966578.09	99				

a. R Squared = .021 (Adjusted R Squared = .011)

There are a number of reasons why the ANOVA on residuals approach does not yield the same results as ANCOVA. I would say that the consensus is that the results from an ANCOVA should be trusted, whereas the ANOVA on residuals should not (García-Berthou, 2001; Maxwell, Delaney, & Manheimer, 1985). Additionally, it should be noted that the ANOVA on residuals approach does not absolve researchers from testing the homogeneity of regression slopes assumption.

Practice Questions

#1. Comparing golf swing training strategies on ball speed.

Had someone told me when I started this book that I would include a study relevant to golf as a practice question, I would have been dubious – but here we are! I found a study that exemplifies the use of a covariate in a manner that increases the effect size, rather than decreases it. Milanese, Corte, Salvetti, Cavedon, and Agostini, (2016) were interested in evaluating two training strategies to help golfers generate more ball speed, when they strike the ball with their golf clubs. One training group ($N = 12$) received the more sophisticated ‘amplification of error method’ (MAE), one training group received a conventional ‘direct instruction’ training experience ($N = 12$), and one control group ($N = 12$) did not receive any instruction. After the training, the three groups were assessed. Specifically, the average speed the 36 participants could generate by striking the ball with their golf club was measured. In addition to ball speed, Milanese et al. (2016) measured each participant’s handicap. A player’s handicap represents their skill at golf. A lower value represents better skill. First, test null the hypothesis that the three groups were associated with equal ball speed means. Then, include the handicap variable as a covariate in the analysis ([Data File: golf_training](#)) (**Watch: Golf Training**).

#2. The effect of inoculations on baby cortisol levels.

Lewis, M., & Thomas, D. (1990). Cortisol release in infants in response to inoculation. *Child Development*, 61(1), 50-59. (**Data File: babies_cortisol**)

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